

Age Problem Revision

Below are the **important formulas** for solving age-related problems, presented with two detailed solved examples.

Key Formulas for Age Problems

1. **Present Age:** Let the present age be x .

2. **Age After n Years:** $x + n$

3. **Age Before n Years:** $x - n$

4. **Ratio-Based Present Age:**

- If the ratio of ages of A and B is $m : n$ and the sum is S , then:

$$\text{A's age} = \frac{m}{m+n} \times S, \quad \text{B's age} = \frac{n}{m+n} \times S$$

5. **Use of Linear Equations:** For future/past ages:

$$\text{A's age after } n \text{ years} = x + n$$

$$\text{B's age after } n \text{ years} = y + n$$

- Set up equations based on problem conditions (sums, differences, multiples, or ratios).

Solved Example 1

Question: The sum of the ages of a father and his son is 60 years. Four years ago, the father's age was three times the age of his son. Find their present ages.

Solution:

Let the present age of the son = x years.

- Father's present age = $60 - x$
- 4 years ago:
 - Son's age = $x - 4$
 - Father's age = $60 - x - 4 = 56 - x$
- According to the question:

$$56 - x = 3(x - 4)$$

$$56 - x = 3x - 12$$

$$56 + 12 = 3x + x$$

$$68 = 4x$$

$$x = 17$$

- Son's present age = **17 years**
- Father's present age = **$60 - 17 = 43$ years**

Solved Example 2

Question: The ratio of the ages of two friends is 4:5. After 6 years, the ratio becomes 5:6. Find their present ages.

Solution:

Let their present ages be $4x$ and $5x$.

- After 6 years:
 - First friend: $4x + 6$
 - Second friend: $5x + 6$
- Given ratio:

$$\frac{4x + 6}{5x + 6} = \frac{5}{6}$$

Cross-multiplied:

$$\begin{aligned}6(4x + 6) &= 5(5x + 6) \\24x + 36 &= 25x + 30 \\36 - 30 &= 25x - 24x \\6 &= x\end{aligned}$$

- Present ages:
 - First friend: $4x = 24$ years
 - Second friend: $5x = 30$ years

These formulas and examples will help you tackle a wide range of **age-based problems** with confidence.